Cal Review Queue — Substrate-CHSH B form disambiguation

Elie (filed at Keeper directive 12:40 EDT afternoon prompt)

2026-05-21 Thursday

Cal Review Queue — Substrate-CHSH B form disambiguation

Background

BST has TWO different Bell-experiment substrate-CHSH predictions currently in materials, with non-equivalent values:

Framing 1: K66 original (Lyra T2399, K52a Sessions 6+ pending)

$$S_{BST}^2 = S_{Tsirelson}^2 \cdot (1 + 1/M_g) = 8 \cdot (1 + 1/127)$$

→ S_BST ≈ 2.8395 (SUPER-Tsirelson by ~0.39%)

Source: K66 substrate-CHSH framework, originally framed by Lyra around Wednesday May 20.

Framing 2: Calibration #17 refinement (Elie Toy 3241 S32, Vol 2 Ch 12 line 64)

$$|S|^2 \leq 126/16 = 7.875$$

(rank-1 max eigenvalue of B^2 with $Tr(B^2) = 126/16$)

→ S_BST ≤ $\sqrt{7.875}$ ≈ 2.8062 (SUB-Tsirelson by ~0.78%)

Source: Calibration #17 rank-1 projector framework (Toy 3241), absorbed into Vol 2 Ch 12 Experimental Program narrative.

The disambiguation question

These two predictions differ in direction (super- vs sub-Tsirelson) and magnitude:

Framing	S_BST	vs Tsirelson $2\sqrt{2} \approx 2.8284$
K66 original	2.8395	+0.39% (super)
Calibration #17	≤ 2.8062	-0.78% (sub)
Tsirelson	2.8284	(QM bound)

Both framings are internally consistent within their derivations. The disambiguation is OPEN.

Impact

Bell experiment falsifier: SP-30 letter dispatch and Vienna/Caltech/Munich/Delft outreach require ONE canonical S_BST prediction. Currently HELD pending disambiguation.

Vol 2 Ch 12 Experimental Program narrative (v0.2 update today) reports BOTH framings honestly per Cal Mode 1 scope; not canonical.

Toy 3257 K52a S39 Bell prediction by candidate (today): used K66 framing 1 for prediction range. Should be revisited after disambiguation.

Resolution path

Lyra Sessions 6+ closure of substrate-CHSH B exact form (multi-month gated) is expected to disambiguate:

- Substrate-CHSH operator B exact form → its eigenstructure → max(|S|) prediction
- If B's spectrum sits at $\pm\sqrt{(126/16)}/2$, then $|S|\_max \leq \sqrt{(126/16)} \approx 2.8062$ (Calibration #17 framing canonical)
- If B's spectrum has additional "phase boost" from substrate cyclotomic structure, super-Tsirelson framing 1 may be canonical

Cal review prompt

For Cal review:

1. Is one framing already canonical and the other deprecated?
2. Are these framings interpretations of the same underlying physics (operator norm vs measurement-expectation), with both correct in their respective senses?
3. If divergent: which is the canonical Bell-experiment prediction for external SP-30 outreach?
4. If divergent: what's the resolution timeline (Lyra Sessions 6+, or independent path)?

Honest scope (Cal Mode 1 register)

Both framings are internally consistent. Disambiguation is not "one is wrong" but rather "which canonical form for external register." Current Vol 2 Ch 12 narrative honestly reports BOTH pending resolution; this is Cal-Mode-1-compliant honest-scope handling pre-closure.

NO resolution attempt by Elie pending Cal review + Lyra Sessions 6+ closure.

— Elie, Cal Review Queue item filed per Keeper directive, 2026-05-21 Thursday 13:02 EDT (actual via date)